

#1

(a)

we have to prove if $\lim_{n \rightarrow \infty} E[(X_n - c)^2] = 0$ then $\forall \varepsilon > 0, \lim_{n \rightarrow \infty} P(|X_n - c| \geq \varepsilon) = 0$

$$P(Y \geq a) \leq \frac{E[Y]}{a}$$

$$\text{令 } Y = (X_n - c)^2, \text{ 則 } Y \geq \varepsilon^2 \Rightarrow |X_n - c| \geq \varepsilon$$

根據 Markov 不等式

$$P(|X_n - c| \geq \varepsilon) = P((X_n - c)^2 \geq \varepsilon^2) \leq \frac{E[(X_n - c)^2]}{\varepsilon^2}, \forall \varepsilon^2 > 0$$

由上式可推得

$$0 \leq \lim_{n \rightarrow \infty} P((X_n - c)^2 \geq \varepsilon^2) \leq \frac{\lim_{n \rightarrow \infty} E[(X_n - c)^2]}{\varepsilon^2}$$

所以當 $\lim_{n \rightarrow \infty} E[(X_n - c)^2] = 0$ 時

$$P((X_n - c)^2 \geq \varepsilon^2) = 0$$

(b)

$$\text{令 } X_n = \begin{cases} 0 & , 1 - \frac{1}{n} \\ n & , \frac{1}{n} \end{cases}$$

$$\text{對任意 } \varepsilon > 0, P(|X_n - 0| \geq \varepsilon) = P(X_n \neq 0) = \frac{1}{n}$$

當 $n \rightarrow \infty$ 時

$$\lim_{n \rightarrow \infty} P(|X_n - 0| \geq \varepsilon) = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$E[(X_n - 0)^2] = n \rightarrow \infty$$

$\Rightarrow$ show that "convergence in probability" does not imply "convergence in the mean square".

#2 we have to prove that if $\forall \varepsilon > 0, \lim_{n \rightarrow \infty} P(|X_n - a| \geq \varepsilon) = 0$
 $\lim_{n \rightarrow \infty} P(|Y_n - b| \geq \varepsilon) = 0$ then $\forall \varepsilon > 0, \lim_{n \rightarrow \infty} P(|Z_n - ab| \geq \varepsilon) = 0$
 $(Z_n = X_n \cdot Y_n)$

$$|Z_n - ab| = |X_n Y_n - ab| = |X_n Y_n - X_n b + X_n b - ab|$$

$$|Z_n - ab| \leq |X_n(Y_n - b)| + |b(X_n - a)|$$

對於 $P(|Z_n - ab| \geq \varepsilon)$, 可分成 $P(|Z_n - ab| \geq \frac{\varepsilon}{2}) \leq P(|X_n||Y_n - b| \geq \frac{\varepsilon}{2})$
 $+ P(|b(X_n - a)| \geq \frac{\varepsilon}{2})$

1° $\because Y_n \rightarrow b$ 且 X_n 有界 ($\lim_{n \rightarrow \infty} X_n = a$)

$$\therefore \lim_{n \rightarrow \infty} P(|X_n||Y_n - b| \geq \frac{\varepsilon}{2}) = 0$$

2° $\because X_n \rightarrow a$ 且 b 是常數

$$\therefore \lim_{n \rightarrow \infty} P(|b(X_n - a)| \geq \frac{\varepsilon}{2}) = 0$$

結合 1° 2° 可知 if $\lim_{n \rightarrow \infty} X_n = a \wedge \lim_{n \rightarrow \infty} Y_n = b$ then $\lim_{n \rightarrow \infty} P(|Z_n - ab| \geq \varepsilon) = 0$

#3

(a)

$$C_{n+1} = C_n \cdot X_{n+1} \Rightarrow E[C_{n+1} | C_n] = E[C_n \cdot X_{n+1} | C_n] = C_n \cdot E[X_{n+1} | C_n]$$

$\because X_n$ 為 i.i.d 且與 C_n 無關

$$\therefore E[C_{n+1} | C_n] = E[X_{n+1}] = 1$$

由以上可知 $E[C_{n+1} | C_n] = C_n \cdot 1 = C_n$ #

c6)

$$C_n = \prod_{k=1}^n X_k \Rightarrow \ln C_n = \sum_{k=1}^n \ln X_k$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \ln X_k = E[\ln X_1] < 0$$

$$\Rightarrow \lim_{n \rightarrow \infty} \ln C_n = \lim_{n \rightarrow \infty} n \cdot \frac{1}{n} \sum_{k=1}^n \ln X_k = -\infty$$

$$\Rightarrow \lim_{n \rightarrow \infty} C_n = e^{\ln C_n} = e^{-\infty} = 0$$

$$\Rightarrow \{C_n\} \text{ 幾乎必然收斂到 } 0, \text{ 即 } P(\lim_{n \rightarrow \infty} C_n = 0) = 1$$

#4

(a)

為了使用 CLT, 考慮標準化的 S_n :

$$Z_n = \frac{S_n - E[S_n]}{\sqrt{\text{Var}(S_n)}} = \frac{S_n - np}{\sqrt{np(1-p)}}$$

當 $n \rightarrow \infty$ 時, Z_n 的 distribution 近似於 $N(0, 1)$, 也就是說

$$\frac{S_n - np}{\sqrt{np(1-p)}} \xrightarrow{d} N(0, 1).$$

因此, 當 n 夠大時, S_n 的分布近似於 $N(np, np(1-p))$

c6)

令 $p = \frac{\lambda}{n}$, $\lambda > 0$ 則 $S_n = \sum_{k=1}^n X_k \sim \text{Binomial}(n, \frac{\lambda}{n})$

$$P(S_n = \lambda) = \binom{n}{\lambda} \left(\frac{\lambda}{n}\right)^\lambda \left(1 - \frac{\lambda}{n}\right)^{n-\lambda}$$

$$\text{當 } n \rightarrow \infty \text{ 時, } \binom{n}{\lambda} \approx \frac{n^\lambda}{\lambda!}, \quad \left(1 - \frac{\lambda}{n}\right)^{n-\lambda} \approx e^{-\lambda}, \quad \left(\frac{\lambda}{n}\right)^\lambda = \frac{\lambda^\lambda}{n^\lambda}$$

$$\Rightarrow \lim_{n \rightarrow \infty} P(S_n = \lambda) \approx \frac{n^\lambda}{\lambda!} \cdot \frac{\lambda^\lambda}{n^\lambda} \cdot e^{-\lambda} = \frac{e^{-\lambda} \lambda^\lambda}{\lambda!} \neq$$

(c)

CLT 假設 p 是常數，而在 (b) 中， $p = \frac{1}{n}$ ，當 $n \rightarrow \infty$ 時 $p \rightarrow 0$ ，使得 $X_i \sim \text{Bernoulli}(p)$ 變得稀疏，因此 S_n 的累積分佈逐漸趨於 Poisson，而非 CLT 所描述的 Normal distribution，所以與 (b) 矛盾

CLT 的正態近似需滿足 n 夠大且 p 不能太小

在 (b) 中

p 隨 n 增大而減小， S_n 中大部分試驗結果為 0，這種稀疏性違反了 CLT 的適用條件，使得 Poisson 成為更準確的近似，所以在 (b) 中 CLT 的效果不佳

#5

(a)

對於畢業生，全兩人均選修 RL 的事件為 A

$$P(A|M=m) = \frac{m}{2n} \cdot \frac{m-1}{2n-1}$$

$$E[X|M=m] = n \cdot P(A|M=m) = n \cdot \frac{m}{2n} \cdot \frac{m-1}{2n-1} = \frac{m(m-1)}{4n-2} \quad \#$$

(b)

$$E[X] = E[E[X|M]] = E\left[\frac{m(m-1)}{4n-2}\right] \quad (\text{將 (a) 的結果代入})$$

由於 $M = \sum_{i=1}^{2n} I_i$ 其中 I_i 為第 i 位學生是否有選修 RL 的變數

($I_i \sim \text{Bernoulli}(p)$)，則：

$$E[M] = E\left[\sum_{i=1}^{2n} I_i\right] = 2np$$

$$E[M(M-1)] = E[M^2] - E[M]$$

⇒ 還需計算 $E[M^2]$

$$M^2 = \left(\sum_{i=1}^{2n} I_i\right)^2$$

$$E[M^2] = E\left[\sum_{i=1}^{2n} I_i^2 + \sum_{i \neq j} I_i I_j\right]$$

$$I_i^2 = I_i \Rightarrow E[I_i^2] = E[I_i] = p$$

$$E[I_i I_j] = E[I_i] E[I_j] = p^2 \quad (\text{因 } I_i, I_j \text{ 獨立})$$

$$\Rightarrow E[M^2] = 2np + 2n(2n-1)p^2$$

$$\text{由上知 } E[M(M-1)] = E[M^2] - E[M] = 2np + 2n(2n-1)p^2 - 2np$$

$$= 2n(2n-1)p^2$$

$$E[X] = E\left[\frac{M(M-1)}{4n-2}\right] = \frac{E[M(M-1)]}{4n-2} = \frac{2n(2n-1)p^2}{4n-2} = np^2 \quad \#$$